

ADRIAN HERRERA

Experience

Interrupt Labs

Principal Vulnerability Researcher

Canberra, Australia

Feb. 2023 – Present

- Develop tools (fuzzers, emulators, static analyzers) for vulnerability research and exploit development
- Software reverse engineering and auditing

Australian National University (ANU)

Adjunct Lecturer

Canberra, Australia

Feb. 2020 – Present

- “Vulnerability Research and Exploit Mitigation” (previously “Software Security”) lecturer, teaching automated vulnerability discovery

Defence Science and Technology Group (DSTG)

Technical Team Lead

Canberra, Australia

Feb. 2021 – Feb. 2023

- Led a team of computer scientists and security researchers undertaking cyber security R&D for the Australian Defence Organisation
- Build and manage relationships with operational clients, international partners, and academia

DSTG

Senior Security Researcher

Canberra, Australia

Aug. 2017 – Feb. 2021

- Research on automated vulnerability discovery (fuzzing, program analysis)

École polytechnique fédérale de Lausanne (EPFL)

Research Engineer (Dependable Systems Lab)

Lausanne, Switzerland

Jul. 2016 – Aug. 2017

- Develop the S²E binary analysis platform

Australian Cyber Security Centre (ACSC)

Malware Analyst and Researcher

Canberra, ACT

Feb. 2015 – Jul. 2016

- Reverse engineering, software development, and research for malware analysis

DSTG

Security Researcher

Adelaide, SA / Canberra, ACT

Jan. 2012 – Jul. 2016

- Research, develop, and deploy malware detection and analysis capabilities for the Australian Cyber Security Centre (ACSC)

Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Summer Intern and Honours Student

Brisbane, QLD

Nov. 2010 – Nov. 2011

- Research on wireless sensor network security

Education

ANU

PhD (Computer Science)

Canberra, ACT

Mar. 2019 – Dec. 2023

- Dissertation: “State Space Search in Fuzzing”

University of Wollongong (UOW)

Bachelor of Engineering (Computer) (Honours I)

Wollongong, NSW

Mar. 2006 – Nov. 2011

Publications

- S. Luo, **A. Herrera**, P. Quirk, M. Chase, D. Ranasinghe, S. Kanhere, “Make out like a (Multi-Armed) Bandit: Improving the Odds of Fuzzer Seed Scheduling with T-SCHEDULER”, *ACM Asia Computer and Communications Security (AsiaCCS)*, 2024
- **A. Herrera**, M. Payer, A. Hosking, “DATAFlow: Toward a Data-Flow-Guided Fuzzer”, *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2023
- Z. Jiang, S. Gan, **A. Herrera**, F. Toffalini, L. Romerio, C. Tang, M. Egele, C. Zhang, M. Payer, “EVOCATIO: Conjuring Bugs from a Single PoC”, *ACM Computer and Communications Security (CCS)*, 2022
- **A. Herrera**, M. Payer, A. Hosking, “Registered Report: DATAFlow – Towards a Data-Flow-Guided Fuzzer”, *Fuzzing Workshop (FUZZING)*, 2022
- **A. Herrera**, H. Gunadi, S. Magrath, M. Norrish, M. Payer, A. Hosking, “Seed Selection for Successful Fuzzing”, *ACM International Symposium on Software Testing and Analysis (ISSTA)*, 2021
- A. Hazimeh, **A. Herrera**, M. Payer, “Magma: A Ground-Truth Fuzzing Benchmark”, *ACM SIGMETRICS*, 2021
- **A. Herrera**, “Optimizing Away JavaScript Obfuscation”, *IEEE Source Code Analysis and Manipulation (SCAM)*, 2020
- **A. Herrera**, H. Gunadi, L. Hayes, S. Magrath, F. Friedlander, M. Sebastian, M. Norrish, A. Hosking, “Corpus Distillation for Effective Fuzzing: A Comparative Evaluation”, *arXiv:1905.13055*, 2019
- B. Zhang, C. Feng, **A. Herrera**, V. Chipounov, G. Candea, “Discover Deeper Bugs with Dynamic Symbolic Execution and Coverage-based Fuzz Testing”, *IET Software*, 2018
- **A. Herrera**, “Automated Analysis of Flash Malware”, *DST Group Report DST-Group-TN-1511*, 2016
- **A. Herrera**, B. Cheney, “JMD: A Hybrid Approach for Detecting Java Malware”, *Australasian Information Security Conference (AISC)*, 2015 (**BEST PAPER AWARD**)
- **A. Herrera**, “How Secure is the Next-Generation Internet? An Examination of IPv6”, *DSTO Report DSTO-GD-0767*, 2013
- **A. Herrera**, W. Hu, “A Key Distribution Protocol for Wireless Sensor Networks”, *IEEE Local Computer Networks (LCN)*, 2012

Talks

- “LLMs and Fuzzing: Payload of the Future or Just Empty Bytes?”, *Australian Reverse Engineering and Vulnerability Research (AUREVR)*, 2025
- “Building Domain-Specific Fuzzers”, *Australian Reverse Engineering and Vulnerability Research (AUREVR)*, 2024
- “The Hitchhiker’s Guide to Fuzzer Coverage Metrics”, *Australasian Software Engineering Summer School (ASESS)*, 2024
- “Code Analysis with Databases”, *CSides Canberra*, 2023
- “Hot Fuzz: We’ve got grey, or... white[-box fuzzers]”, *Australian Reverse Engineering and Vulnerability Research (AUREVR)*, 2022
- “Seed Selection for Successful Fuzzing”, *CSIRO’s Data61 & DST Cyber Security Summer School (CSSS)*, 2022
- “Analyzing Trigger-Based Malware with S²E”, *Malware Reverse Engineering (MRE)*, 2019
- “Deobfuscating JavaScript Malware”, *BSides Canberra*, 2019 (**BEST SPEAKER AWARD**)
- “JavaScript Obfuscation”, *CSides Canberra*, 2018
- “Program Analysis for Reverse Engineers: from $\top$ to $\perp$ ”, *BSides Canberra*, 2018

Service

- Program committee, *Source Code Analysis and Manipulation* (SCAM) engineering track, 2025
- Artifact evaluation committee, *Usenix Security* (SEC), 2025 (**NOTEWORTHY REVIEWER AWARD**)
- Artifact evaluation committee, *Workshop on Offensive Technologies* (WOOT), 2023, 2024
- Technical reviewer, *Exploring and fuzzing IoTs Firmware with Qemu* book, 2022
- Artifact evaluation committee, *Fuzzing Workshop* (FUZZING), 2022
- Organizing committee, *CSIRO's Data61 & DSTG Cyber Security Summer School* (CSSS), 2022
- Organizing committee, *International Summer School on Information Security and Protection* (ISSISP), 2018